Equity Portfolio Optimisation with Net Zero Objectives

Exercise 3 — Mathematical Report

Green Finance: Risk and Protfolio Management
2025 – 2026

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Question 1 — Benchmark Analysis

1.1 Risk Model

Under the Capital Asset Pricing Model (CAPM) each sector return satisfies

$$R_i = r + \beta_i\,(R_m - r) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \tilde{\sigma}_i^2), \quad R_m \sim \mathcal{N}(\mu_m, \sigma_m^2), \tag{1}$$

with $r = 3\%$, $\sigma_m = 20\%$, and all ε_i mutually independent and independent of R_m.

1.2 (a) — Covariance Matrix, Correlation Matrix and Sector Volatilities

1.2.1 Derivation

From (1), the covariance between sectors i and j is

$$\begin{aligned}\Sigma_{ij} &= \mathrm{Cov}(R_i, R_j)\\ &= \mathrm{Cov}\big(\beta_i(R_m - r) + \varepsilon_i,\ \beta_j(R_m - r) + \varepsilon_j\big)\\ &= \beta_i\beta_j \mathrm{Var}(R_m - r) + \mathrm{Cov}(\varepsilon_i, \varepsilon_j)\\ &= \beta_i\beta_j\,\sigma_m^2 + \delta_{ij}\,\tilde{\sigma}_i^2.\end{aligned} \tag{2}$$

where δ_{ij} is the Kronecker delta, and where the idiosyncratic shocks ε_i are independent of R_m and mutually independent across sectors. In matrix form:

$$\boxed{\boldsymbol{\Sigma} = \sigma_m^2\,\boldsymbol{\beta}\boldsymbol{\beta}^\top + \tilde{\mathbf{D}}^2} \tag{3}$$

with $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_{11})^\top$ and $\tilde{\mathbf{D}} = \mathrm{diag}(\tilde{\sigma}_1, \ldots, \tilde{\sigma}_{11})$.

The sector volatility and correlation are then

$$\sigma_i = \sqrt{\Sigma_{ii}} = \sqrt{\beta_i^2\sigma_m^2 + \tilde{\sigma}_i^2}, \qquad \rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i\sigma_j}. \tag{4}$$

1.2.2 Numerical Results

Table 1: Sector volatilities σ_i

Sector	β_i	$\tilde{\sigma}_i$ (%)	σ_i (%)
Communication Services	0.95	26.2	32.36
Consumer Discretionary	1.05	32.9	39.03
Consumer Staples	0.45	21.1	22.94
Energy	1.40	33.8	43.89
Financials	1.15	23.1	32.60
Health Care	0.75	25.9	29.93
Industrials	1.00	26.5	33.20
Information Technology	1.20	27.1	36.20
Materials	1.10	30.1	37.28
Real Estate	0.80	27.4	31.73
Utilities	0.70	22.8	26.76

$$\Sigma(\%) = \begin{pmatrix} 10.47 & 3.99 & 1.71 & 5.32 & 4.37 & 2.85 & 3.80 & 4.56 & 4.18 & 3.04 & 2.66 \\ 3.99 & 15.23 & 1.89 & 5.88 & 4.83 & 3.15 & 4.20 & 5.04 & 4.62 & 3.36 & 2.94 \\ 1.71 & 1.89 & 5.26 & 2.52 & 2.07 & 1.35 & 1.80 & 2.16 & 1.98 & 1.44 & 1.26 \\ 5.32 & 5.88 & 2.52 & 19.26 & 6.44 & 4.20 & 5.60 & 6.72 & 6.16 & 4.48 & 3.92 \\ 4.37 & 4.83 & 2.07 & 6.44 & 10.63 & 3.45 & 4.60 & 5.52 & 5.06 & 3.68 & 3.22 \\ 2.85 & 3.15 & 1.35 & 4.20 & 3.45 & 8.96 & 3.00 & 3.60 & 3.30 & 2.40 & 2.10 \\ 3.80 & 4.20 & 1.80 & 5.60 & 4.60 & 3.00 & 11.02 & 4.80 & 4.40 & 3.20 & 2.80 \\ 4.56 & 5.04 & 2.16 & 6.72 & 5.52 & 3.60 & 4.80 & 13.10 & 5.28 & 3.84 & 3.36 \\ 4.18 & 4.62 & 1.98 & 6.16 & 5.06 & 3.30 & 4.40 & 5.28 & 13.90 & 3.52 & 3.08 \\ 3.04 & 3.36 & 1.44 & 4.48 & 3.68 & 2.40 & 3.20 & 3.84 & 3.52 & 10.07 & 2.24 \\ 2.66 & 2.94 & 1.26 & 3.92 & 3.22 & 2.10 & 2.80 & 3.36 & 3.08 & 2.24 & 7.16 \end{pmatrix} \tag{5}$$

1.3 (b) — Benchmark Volatility

1.3.1 Derivation

For any portfolio $\mathbf{w}$:

$$\sigma^2(\mathbf{w}) = \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w} = \sigma_m^2\, (\boldsymbol{\beta}^\top \mathbf{w})^2 + \sum_{i=1}^{11} w_i^2\, \tilde{\sigma}_i^2. \tag{6}$$

1.3.2 Numerical Results

$$\sigma(\mathbf{b}) = \sqrt{\mathbf{b}^\top \boldsymbol{\Sigma}\, \mathbf{b}} = \mathbf{22.21}\% \qquad \text{vs} \qquad \sigma_m = 20\%.$$

Comments

$\sigma(\mathbf{b}) \neq \sigma_m$ because the benchmark $\mathbf{b}$ is *not* the true market portfolio. Under CAPM, the market portfolio is capitalization-weighted across *all* assets; the idiosyncratic terms cancel completely in the limit. Here the benchmark covers only 11 synthetic sector assets with rounded weights, so:

- Residual idiosyncratic variance $\sum_i b_i^2 \tilde{\sigma}_i^2 > 0$ persists.
- The sector weights differ from true float-adjusted market-cap weights.

The benchmark volatility therefore slightly exceeds σ_m.

1.4 (c) — Weighted-Average Beta

1.4.1 Derivation

The portfolio beta is linear in weights:

$$\beta(\mathbf{b}) = \sum_{i=1}^{11} b_i \beta_i = \mathbf{b}^\top \boldsymbol{\beta}. \tag{7}$$

For the *true* market portfolio the result would be exactly 1 by construction.

1.4.2 Numerical Results

$$\beta(\mathbf{b}) = \sum_{i=1}^{11} b_i \beta_i = \mathbf{1.0001} \approx 1.$$

Comments

The result is very close to 1 because the benchmark is a broad market proxy. The tiny deviation from 1 arises from rounding in the published benchmark weights (the b_i sum to exactly 100% but the betas

are not perfectly calibrated to a 100% market-cap universe).

1.5 (d) — Implied Risk Premia and Expected Returns

1.5.1 Derivation

The Sharpe ratio of any portfolio $\mathbf{w}$ is defined as

$$\mathrm{SR}(\mathbf{w} \mid r) = \frac{\mu(\mathbf{w}) - r}{\sigma(\mathbf{w})} = 0.25. \tag{8}$$

For the market portfolio specifically:

$$\mu_m - r = \mathrm{SR} \cdot \sigma_m = 0.25 \times 0.20 = 5\%. \tag{9}$$

Under CAPM, the *implied* (market-priced) expected excess return of sector i is

$$\boxed{\tilde{\pi}_i = \beta_i \, (\mu_m - r), \qquad \tilde{\mu}_i = r + \tilde{\pi}_i.} \tag{10}$$

1.5.2 Numerical Results

Table 2: Implied risk premia and expected returns

Sector	β_i	$\tilde{\pi}_i$ (%)	$\tilde{\mu}_i$ (%)
Communication Services	0.95	4.75	7.75
Consumer Discretionary	1.05	5.25	8.25
Consumer Staples	0.45	2.25	5.25
Energy	1.40	7.00	10.00
Financials	1.15	5.75	8.75
Health Care	0.75	3.75	6.75
Industrials	1.00	5.00	8.00
Information Technology	1.20	6.00	9.00
Materials	1.10	5.50	8.50
Real Estate	0.80	4.00	7.00
Utilities	0.70	3.50	6.50

1.6 (e) — Climate Metrics of the Benchmark

1.6.1 Derivation

All climate metrics are defined as weighted averages:

$$\mathrm{CI}(\mathbf{w}) = \sum_{i=1}^{11} w_i \, \mathrm{CI}_i, \quad \mathrm{CM}(\mathbf{w}) = \sum_{i=1}^{11} w_i \, \mathrm{CM}_i, \quad \mathrm{GI}(\mathbf{w}) = \sum_{i=1}^{11} w_i \, \mathrm{GI}_i. \tag{11}$$

1.6.2 Numerical Results

$$\begin{aligned}
\mathrm{CI}(\mathbf{b})_{\mathrm{Sc1+2}} &= \mathbf{127.07}\ \mathrm{tCO_2e/\$mn} \\
\mathrm{CI}(\mathbf{b})_{\mathrm{Sc1+2+3}} &= \mathbf{267.01}\ \mathrm{tCO_2e/\$mn} \\
\mathrm{CM}(\mathbf{b})_{\mathrm{Sc1+2}} &= \mathbf{-5.93}\% \\
\mathrm{CM}(\mathbf{b})_{\mathrm{Sc1+2+3}} &= \mathbf{-0.95}\% \\
\mathrm{GI}(\mathbf{b}) &= \mathbf{0.84}\%
\end{aligned}$$

Question 2 — CTB Decarbonisation Pathway

2.1 (a) — Optimisation Problem and QP Form

2.1.1 Problem Statement

The investor minimises tracking-error volatility subject to a decarbonisation constraint following the Climate Transition Benchmark (CTB) trajectory:

$$\mathrm{CI}^\star(t) = (1 - 30\%)(1 - 7\%)^t \,\mathrm{CI}(\mathbf{b}). \tag{12}$$

The optimisation problem is:

$$\mathbf{w}^\star(t) = \arg\min_{\mathbf{w}} (\mathbf{w} - \mathbf{b})^\top \boldsymbol{\Sigma} (\mathbf{w} - \mathbf{b}) \tag{13}$$

subject to:

$$\mathbf{1}^\top \mathbf{w} = 1 \qquad \text{(fully invested)} \tag{14}$$

$$w_i \geq 0 \qquad \text{(long only)} \tag{15}$$

$$\mathbf{CI}^\top \mathbf{w} \leq \mathrm{CI}^\star(t) \qquad \text{(CTB decarbonisation)} \tag{16}$$

2.1.2 QP Form

Expanding the objective:

$$(\mathbf{w} - \mathbf{b})^\top \boldsymbol{\Sigma} (\mathbf{w} - \mathbf{b}) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} - 2\mathbf{b}^\top \boldsymbol{\Sigma} \mathbf{w} + \underbrace{\mathbf{b}^\top \boldsymbol{\Sigma} \mathbf{b}}_{\text{constant}}.$$

This gives the standard QP:

QP form

$$\min_{\mathbf{w}} \ \tfrac{1}{2} \mathbf{w}^\top Q \, \mathbf{w} + \mathbf{c}^\top \mathbf{w} \quad \text{with} \quad Q = 2\boldsymbol{\Sigma}, \quad \mathbf{c} = -2\boldsymbol{\Sigma}\mathbf{b}$$

Linear constraints:

$$(\mathbf{1}^\top)\mathbf{w} = 1, \qquad \mathbf{CI}^\top \mathbf{w} \leq \mathrm{CI}^\star(t), \qquad \mathbf{0} \leq \mathbf{w} \leq \mathbf{1}.$$

2.2 (b) — Optimised Portfolios: Scope 1+2

Table 3: Portfolio statistics at each rebalancing date (Scope 1+2)

t	$\mathrm{CI}^\star(t)$	TE vol (%)	$\mathrm{CI}(\mathbf{w})$	$\mathrm{CM}(\mathbf{w})$ (%)	$\mathrm{GI}(\mathbf{w})$ (%)	$R(t, \mathbf{w})$ (%)
0	88.95	0.58	88.95	−5.82	0.66	30.00
1	82.72	0.67	82.72	−5.81	0.63	34.90
2	76.93	0.76	76.93	−5.79	0.61	39.46
5	61.88	1.13	61.88	−5.79	0.57	51.30
10	43.05	1.96	43.05	−5.81	0.55	66.12

Table 4: Optimal weights $\mathbf{w}^\star(t)$ (%) — Scope 1+2

Sector	$t=0$	$t=1$	$t=2$	$t=5$	$t=10$
Communication Services	8.54	8.60	8.65	9.07	9.73
Consumer Discretionary	12.48	12.51	12.54	12.74	13.05
Consumer Staples	7.48	7.58	7.67	7.94	8.30
Energy	2.95	2.93	2.91	2.12	0.78
Financials	13.61	13.68	13.74	14.38	15.41
Health Care	12.98	13.04	13.10	13.48	14.08
Industrials	10.42	10.46	10.50	10.56	10.63
Information Technology	23.29	23.33	23.38	23.84	24.58
Materials	4.23	4.18	4.14	2.66	0.13
Real Estate	3.05	3.10	3.13	3.21	3.30
Utilities	0.96	0.59	0.25	0.00	0.00

Comments

- As t increases, $\mathrm{CI}^\star(t)$ tightens, forcing larger underweights in high-carbon sectors (Energy CI = 434, Materials CI = 559, Utilities CI = 1655). Utilities is eliminated by $t=5$.
- Tracking error grows monotonically with t: the portfolio must deviate further from the benchmark to meet the tightening CI ceiling.
- The reduction rate $R(t,\mathbf{w}) = 1 - \mathrm{CI}(\mathbf{w})/\mathrm{CI}(\mathbf{b})$ meets the CTB target exactly at each date.
- Carbon momentum $\mathrm{CM}(\mathbf{w})$ remains approximately stable around -5.8%, slightly better than the benchmark (-5.93%).

2.3 (c) — Optimised Portfolios: Scope 1+2+3 Upstream

The same QP is solved but replacing $\mathbf{CI}_{\mathrm{Sc1+2}}$ with $\mathbf{CI}_{\mathrm{Sc1+2+3}}$ and computing $\mathrm{CI}^\star(t) = (1-30\%)(1-7\%)^t \cdot \mathrm{CI}(\mathbf{b})_{\mathrm{Sc1+2+3}}$.

Table 5: Portfolio statistics (Scope 1+2+3 upstream)

t	$\mathrm{CI}^\star(t)$	TE vol (%)	$\mathrm{CI}(\mathbf{w})$	$\mathrm{CM}(\mathbf{w})$ (%)	$\mathrm{GI}(\mathbf{w})$ (%)	$R(t,\mathbf{w})$ (%)
0	186.91	1.23	186.91	−0.73	0.58	30.00
1	173.82	1.59	173.82	−0.71	0.57	34.90
2	161.66	1.94	161.66	−0.70	0.56	39.46
5	130.03	3.27	130.03	−0.71	0.52	51.30
10	90.46	8.17	90.46	−0.68	0.10	66.12

Comments

- Including Scope 3 upstream emissions dramatically raises the CI of Consumer Staples (CI = 392) and Consumer Discretionary (CI = 203), sectors with large supply-chain footprints.
- The tracking error is substantially higher at every horizon compared to Scope 1+2 only.
- At $t=10$, the TE vol exceeds 8%, making pure passive ESG management very costly.
- Utilities is zeroed out immediately at $t=0$ (its Scope 3 CI of 1867 is prohibitive).

2.4 (d) — Implied Bets

2.4.1 Derivation

If portfolio $\mathbf{w}$ is *mean–variance optimal*, its implied excess returns satisfy:

$$\tilde{\boldsymbol{\mu}}(\mathbf{w}) - r\mathbf{1} = \lambda\,\boldsymbol{\Sigma}\mathbf{w}, \qquad \lambda = \frac{\mathrm{SR}(\mathbf{w})}{\sigma(\mathbf{w})}. \tag{17}$$

The *implied bet* of sector i is the difference between the return priced in by $\mathbf{w}^\star(t)$ and the market-implied return:

$$\boxed{\Delta\tilde{\mu}_i(\mathbf{w}^\star(t)) = \tilde{\mu}_i(\mathbf{w}^\star(t)) - \tilde{\mu}_i} \tag{18}$$

The derivation of equation (17) is given in Appendix A.

2.4.2 Numerical Results

Table 6: Implied bets $\Delta\tilde{\mu}_i(\mathbf{w}^\star)$ in % — Scope 1+2

Sector	$t=0$	$t=1$	$t=2$	$t=5$	$t=10$
Communication Services	+0.011	+0.013	+0.014	+0.029	+0.045
Consumer Discretionary	+0.002	+0.002	+0.002	+0.007	+0.006
Consumer Staples	+0.022	+0.025	+0.028	+0.034	+0.036
Energy	−0.037	−0.044	−0.050	−0.176	−0.400
Financials	+0.007	+0.007	+0.008	+0.025	+0.045
Health Care	+0.015	+0.017	+0.019	+0.034	+0.052
Industrials	+0.002	+0.002	+0.001	−0.012	−0.043
Information Technology	−0.000	−0.001	−0.001	+0.015	+0.033
Materials	−0.043	−0.051	−0.058	−0.228	−0.525
Real Estate	+0.010	+0.012	+0.013	+0.005	−0.017
Utilities	−0.140	−0.163	−0.184	−0.212	−0.238

Table 7: Implied bets $\Delta\tilde{\mu}_i(\mathbf{w}^\star)$ in % — Scope 1+2+3 upstream

Sector	$t=0$	$t=1$	$t=2$	$t=5$	$t=10$
Communication Services	+0.053	+0.070	+0.085	+0.205	+0.637
Consumer Discretionary	−0.004	−0.009	−0.015	−0.125	−1.586
Consumer Staples	−0.028	−0.055	−0.080	−0.362	−0.442
Energy	−0.216	−0.315	−0.408	−0.523	−0.699
Financials	+0.050	+0.069	+0.084	+0.207	+0.924
Health Care	+0.044	+0.055	+0.064	+0.131	−0.082
Industrials	−0.022	−0.037	−0.053	−0.257	−1.020
Information Technology	+0.012	+0.018	+0.021	+0.000	−0.342
Materials	−0.233	−0.343	−0.446	−0.556	−0.695
Real Estate	+0.045	+0.055	+0.064	+0.129	−0.222
Utilities	−0.212	−0.221	−0.231	−0.249	−0.337

Comments

- **Negative bets** on Energy, Materials and Utilities reflect the implicit view that these sectors are *overpriced* relative to what the market prices in — the investor systematically underweights them to meet the CI constraint. Under Scope 1+2, Utilities carries the largest negative bet (−0.140% at $t=0$, −0.238% at $t=10$) because it has by far the highest carbon intensity (CI = 1655 tCO_2e/\$mn). Under Scope 1+2+3, Energy and Materials are also heavily penalized, since their upstream emissions are very high (CI = 803 and CI = 892 respectively).
- **The magnitude of bets grows with** t as the threshold $\mathrm{CI}^\star(t)$ tightens by 7% per year, forcing the optimizer to deviate increasingly from the benchmark. Under Scope 1+2+3 this growth is far more dramatic: by $t=10$, Consumer Discretionary reaches −1.586% and Industrials −1.020%, sectors that were barely affected under Scope 1+2, because their upstream supply-chain emissions are large.
- **Positive bets** on Communication Services, Financials and Health Care arise because the full-investment constraint $\sum_i w_i = 1$ forces overweights in low-carbon sectors to compensate for the underweights. Under Scope 1+2+3, these positive bets are much larger (e.g. Financials reaches +0.924% at $t=10$) because more sectors must be heavily reduced, concentrating the offsetting overweights in fewer places.

2.5 (e) — Expected Carbon Intensity Before Rebalancing at $t=1$

2.5.1 Derivation

The investor holds $\mathbf{w}^\star(0)$ from $t=0$ to just before $t=1$. Over one year each sector's carbon intensity evolves according to its carbon momentum:

$$\mathrm{CI}_i(1) = \mathrm{CI}_i(0)\,(1+\mathrm{CM}_i). \tag{19}$$

The portfolio CI just before rebalancing at $t = 1$ is therefore:

$$\boxed{\mathrm{CI}(\mathbf{w}^\star(0),\, t = 1^-) = \sum_{i=1}^{11} w_i^\star(0)\, \mathrm{CI}_i(0)\, (1 + \mathrm{CM}_i).} \tag{20}$$

This is compared with the threshold $\mathrm{CI}^\star(1) = (1 - 30\%)(1 - 7\%)\, \mathrm{CI}(\mathbf{b})$.

2.5.2 Numerical Results

$$\begin{aligned}
\mathrm{CI}(\mathbf{w}^\star(0)) &= 88.95\ \mathrm{tCO_2e/\$mn} \\
\mathrm{CI}(\mathbf{w}^\star(0),\, t = 1^-) &= \mathbf{83.24}\ \mathrm{tCO_2e/\$mn} \\
\mathrm{CI}^\star(1) &= 82.72\ \mathrm{tCO_2e/\$mn} \\
\text{Shortfall} &= +0.52\ \mathrm{tCO_2e/\$mn} \quad \Rightarrow \text{ constraint violated}
\end{aligned}$$

Comments

Even though $\mathbf{w}^\star(0)$ exactly satisfies the $t = 0$ constraint, by $t = 1$ the portfolio CI has only declined by the *portfolio-average* carbon momentum ($\approx -5.8\%$), whereas the CTB pathway requires a -7% annual reduction. The shortfall of 0.52 $\mathrm{tCO_2e/\$mn}$ means that **annual rebalancing is mandatory** to stay on the CTB trajectory.

Question 3 — Additional Constraints (Scope 1+2, $t = 0$)

Unless stated otherwise, all portfolios in this section use Scope 1+2 emissions, $t = 0$, and include the baseline CTB constraint $\mathrm{CI}(\mathbf{w}) \leq \mathrm{CI}^\star(0) = 88.95$.

3.1 (a) — Minimum-Weight Constraint

3.1.1 Constraint Interpretation

The constraint

$$w_i \geq \frac{b_i}{2} \quad \forall i \tag{21}$$

prevents extreme underweighting of any sector. It limits concentration risk and ensures some exposure to every benchmark constituent, even in a heavily carbon-constrained portfolio.

3.1.2 QP Form

The modification is simply a change in the lower bound of $\mathbf{w}$:

QP — constraint Form

$$\min_{\mathbf{w}}\ \tfrac{1}{2}\mathbf{w}^\top Q\, \mathbf{w} + \mathbf{c}^\top \mathbf{w}$$

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{CI}^\top \mathbf{w} \leq \mathrm{CI}^\star(0), \quad \tfrac{1}{2}\mathbf{b} \leq \mathbf{w} \leq \mathbf{1}.$$

3.1.3 Results

Table 8: Optimal portfolio — minimum-weight constraint

Metric	Value
TE vol	0.80%
CI($\mathbf{w}$)	88.95 tCO_2e/\$mn
$R(t, \mathbf{w})$	30.00%
CM($\mathbf{w}$)	−5.87%
GI($\mathbf{w}$)	0.70%

Sector	$w_i^\star$ (%)
Communication Services	8.83
Consumer Discretionary	12.61
Consumer Staples	7.53
Energy	2.22
Financials	14.09
Health Care	13.22
Industrials	10.40
Information Technology	23.64
Materials	2.84
Real Estate	3.03
Utilities	1.60

Comments

The floor prevents zeroing out any sector: Utilities retains 1.60% (vs 0.96% baseline), Energy 2.22% (vs 2.95%), Materials 2.84% (vs 4.23%). The 30% CI reduction is still achieved. The TE vol of 0.80% is only slightly higher than the unconstrained 0.58%, making this a practically appealing constraint.

3.2 (b) — Carbon-Momentum Constraint

3.2.1 Economic Interpretation

The constraint

$$\mathrm{CM}(\mathbf{w}) = \sum_{i=1}^{11} w_i \, \mathrm{CM}_i \leq \mathrm{CM}^\star \tag{22}$$

ensures the portfolio tilts toward *fast-decarbonising* sectors, not merely *currently low-carbon* ones. This guards against the "low-carbon trap" where a portfolio holds emitters with slow improvement trajectories.

3.2.2 QP Form

QP — constraint (22)

$$\min_{\mathbf{w}} \ \tfrac{1}{2}\mathbf{w}^\top Q\,\mathbf{w} + \mathbf{c}^\top \mathbf{w}$$

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \begin{pmatrix} \mathbf{CI}^\top \\ \mathbf{CM}^\top \end{pmatrix} \mathbf{w} \leq \begin{pmatrix} \mathrm{CI}^\star(0) \\ \mathrm{CM}^\star \end{pmatrix}, \quad \mathbf{0} \leq \mathbf{w} \leq \mathbf{1}.$$

3.2.3 Results

Table 9: Results by $\mathrm{CM}^\star$ threshold

$\mathrm{CM}^\star$	TE vol (%)	CI($\mathbf{w}$)	R(%)	CM($\mathbf{w}$) (%)	GI($\mathbf{w}$) (%)
−5%	0.58	88.95	30.0	−5.82	0.66
−6%	0.79	88.95	30.0	−6.00	0.68
−7%	3.70	88.95	30.0	−7.00	0.81
−8%	9.04	88.95	30.0	−8.00	1.04

Table 10: Optimal weights $\mathbf{w}^\star$ (%) by $\mathrm{CM}^\star$ threshold

Sector	$\mathrm{CM}^\star = -5\%$	$\mathrm{CM}^\star = -6\%$	$\mathrm{CM}^\star = -7\%$	$\mathrm{CM}^\star = -8\%$
Communication Services	8.54	8.02	3.79	0.00
Consumer Discretionary	12.48	12.79	14.55	14.88
Consumer Staples	7.48	6.44	0.00	0.00
Energy	2.95	2.24	0.00	0.00
Financials	13.61	14.57	20.50	32.06
Health Care	12.98	13.70	18.20	22.85
Industrials	10.42	11.17	15.84	24.75
Information Technology	23.29	23.09	21.10	2.84
Materials	4.23	4.31	4.45	0.01
Real Estate	3.05	2.55	0.00	0.00
Utilities	0.96	1.11	1.58	2.62

Comments

The benchmark CM is -5.93%, so $\mathrm{CM}^\star = -5\%$ is non-binding. At $\mathrm{CM}^\star = -6\%$ the constraint is mildly binding (TE $= 0.79\%$). At $\mathrm{CM}^\star = -7\%$ the TE jumps to 3.70% as the optimizer is forced into Industrials and Health Care (fast decarbonisers). At $\mathrm{CM}^\star = -8\%$ the solution becomes very concentrated (TE $= 9\%$).

3.3 (c) — Green-Intensity Constraint

3.3.1 Economic Interpretation

The constraint

$$\mathrm{GI}(\mathbf{w}) = \sum_{i=1}^{11} w_i \, \mathrm{GI}_i \geq (1+G)\,\mathrm{GI}(\mathbf{b}) \tag{23}$$

is a *positive screening* constraint requiring the portfolio to hold proportionally more revenues aligned with the EU Taxonomy than the benchmark. $G = 0$ means at least matching the benchmark; $G = 2$ means tripling it.

3.3.2 QP Form

QP Form

$$\min_{\mathbf{w}} \ \tfrac{1}{2}\mathbf{w}^\top Q\,\mathbf{w} + \mathbf{c}^\top \mathbf{w}$$

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{CI}^\top \mathbf{w} \leq \mathrm{CI}^\star(0), \quad -\mathbf{GI}^\top \mathbf{w} \leq -(1+G)\mathrm{GI}(\mathbf{b}), \quad \mathbf{0} \leq \mathbf{w} \leq \mathbf{1}.$$

3.3.3 Results

Table 11: Results by green-intensity multiplier G

G	GI target (%)	TE vol (%)	$\mathrm{CI}(\mathbf{w})$	$R(\%)$	$\mathrm{GI}(\mathbf{w})$ (%)
0.0	0.84	1.64	88.95	30.0	0.84
0.5	1.26	5.92	88.95	30.0	1.26
1.0	1.68	11.26	88.95	30.0	1.68
2.0	2.52		Infeasible		

Table 12: Optimal weights $\mathbf{w}^\star$ (%) by green-intensity multiplier G

Sector	$G = 0$	$G = 0.5$	$G = 1$	$G = 2$
Communication Services	8.06	4.42	0.00	—
Consumer Discretionary	13.86	18.67	24.02	—
Consumer Staples	6.14	0.00	0.00	—
Energy	1.35	0.00	0.00	—
Financials	13.15	8.46	0.00	—
Health Care	12.44	9.07	1.76	—
Industrials	13.77	26.55	42.17	—
Information Technology	23.29	21.25	16.75	—
Materials	1.41	0.00	0.00	—
Real Estate	4.49	9.59	14.40	—
Utilities	2.05	1.99	0.89	—

$G = 2$: infeasible — no solution exists satisfying all constraints simultaneously.

Comments

Utilities has the highest GI (8.4%) but also the highest CI (1655 tCO_2e/\$mn), creating a direct tension with the CI constraint. At $G = 2$ (tripling benchmark GI to 2.52%), the problem becomes **infeasible**: there is not enough green-revenue capacity in the universe to achieve this target while simultaneously meeting the 30% CI reduction. Industrials (GI=2.4%) absorbs most of the weight as G increases.

3.4 (d) — Combined Constraints

3.4.1 QP Form

QP — all three constraints combined

$$\min_{\mathbf{w}} \ \tfrac{1}{2}\mathbf{w}^\top Q\,\mathbf{w} + \mathbf{c}^\top \mathbf{w}$$

$$\mathbf{1}^\top \mathbf{w} = 1, \qquad \begin{pmatrix} \mathbf{CI}^\top \\ \mathbf{CM}^\top \\ -\mathbf{GI}^\top \end{pmatrix} \mathbf{w} \le \begin{pmatrix} \mathrm{CI}^\star(0) \\ \mathrm{CM}^\star \\ -(1+G)\mathrm{GI}(\mathbf{b}) \end{pmatrix}, \qquad \tfrac{1}{2}\mathbf{b} \le \mathbf{w} \le \mathbf{1}.$$

3.4.2 Results

Table 13: Results for combined-constraint scenarios

Scenario	TE vol (%)	CI	R(%)	CM(%)	GI(%)
$\mathrm{CM}^\star = -6\%,\ G = 0.25$	9.49	88.95	30.0	−6.27	1.05
$\mathrm{CM}^\star = 0\%,\ G = 0.50$			Infeasible		
$\mathrm{CM}^\star = -7\%,\ G = 0.00$	5.03	88.95	30.0	−7.00	0.86
$\mathrm{CM}^\star = -7\%,\ G = 0.25$			Infeasible		

3.5 (e) — Comment and Realism Assessment

Table 14: Summary of all $t = 0$ Scope 1+2 solutions

Scenario	TE vol (%)	CI	R(%)	CM(%)	GI(%)
Q2(b) baseline	0.58	88.95	30.0	−5.82	0.66
3(a) $w_i \geq b_i/2$	0.80	88.95	30.0	−5.87	0.70
3(b) $\text{CM}^\star = -6\%$	0.79	88.95	30.0	−6.00	0.68
3(b) $\text{CM}^\star = -7\%$	3.70	88.95	30.0	−7.00	0.81
3(c) $G = 0.5$	5.92	88.95	30.0	−6.50	1.26
3(c) $G = 1.0$	11.26	88.95	30.0	−6.65	1.68
3(d) $\text{CM}^\star = -6\%, G = 0.25$	9.49	88.95	30.0	−6.27	1.05
3(d) $\text{CM}^\star = -7\%, G = 0.00$	5.03	88.95	30.0	−7.00	0.86

Comments

Realistic solutions (low-to-moderate TE, feasible, well-diversified):

- **Q2(b) baseline**: The minimum viable ESG approach. TE = 0.58%, 30% CI reduction, no additional friction.
- **3(a)** $w_i \geq b_i/2$: Adds a sensible diversification floor at negligible TE cost (+0.22pp). Practically implementable for risk-averse mandates.
- **3(b)** $\text{CM}^\star = -6\%$: Mildly binding momentum tilt, TE = 0.79%. Adds a forward-looking dimension without distorting the portfolio.

Problematic / unrealistic solutions:

- **3(b)** $\text{CM}^\star \in \{-7\%, -8\%\}$: TE spikes to 3.7%–9%; heavy concentration in Industrials and Health Care.
- **3(c)** $G \geq 0.5$: Large TE (5.9%+) and at $G = 2$ infeasible due to the conflict between green revenues (Utilities) and carbon reduction.
- **3(d)** $\text{CM}^\star = 0\%$: The momentum constraint is vacuous since $\text{CM}(\mathbf{b}) < 0$; the solution degenerates to the GI-only case. At $\text{CM}^\star = -7\%, G = 0.25$ the problem is infeasible.

Conclusion: The green-intensity constraint (EU Taxonomy alignment) and the carbon-intensity constraint are not always aligned. Utilities, the sector with the highest green revenue share (8.4%), is also the most carbon-intensive (CI = 1655 $\text{tCO}_2\text{e}/\$\text{mn}$). Investors must recognise that *being green* and *being low-carbon* are complementary but distinct objectives that can conflict in practice.

Proof: Implied Returns from the Mean-Variance FOC

Step 1 — Mean-variance optimality conditions

A mean-variance investor solves:

$$\max_{\mathbf{w}} \ \mathbf{w}^\top \boldsymbol{\mu} - \frac{\lambda}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1. \tag{24}$$

Forming the Lagrangian with multiplier γ on the budget constraint:

$$\mathcal{L} = \mathbf{w}^\top \boldsymbol{\mu} - \frac{\lambda}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w} - \gamma(\mathbf{1}^\top \mathbf{w} - 1), \tag{25}$$

the first-order condition (FOC) is:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \boldsymbol{\mu} - \lambda \boldsymbol{\Sigma}\mathbf{w} - \gamma \mathbf{1} = \mathbf{0} \quad \Rightarrow \quad \boldsymbol{\mu} = \gamma \mathbf{1} + \lambda \boldsymbol{\Sigma}\mathbf{w}. \tag{26}$$

Step 2 — Identifying the Lagrange multiplier $\gamma = r$

Pre-multiplying (26) by $\mathbf{w}^\top$:

$$\mu(\mathbf{w}) = \gamma \underbrace{\mathbf{1}^\top \mathbf{w}}_{=1} + \lambda \underbrace{\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w}}_{=\sigma^2(\mathbf{w})} \quad \Rightarrow \quad \mu(\mathbf{w}) = \gamma + \lambda \sigma^2(\mathbf{w}). \tag{27}$$

The Sharpe ratio condition $\mathrm{SR}(\mathbf{w}) = \frac{\mu(\mathbf{w}) - r}{\sigma(\mathbf{w})}$ gives:

$$\mathrm{SR}(\mathbf{w}) = \frac{\gamma - r + \lambda \sigma^2(\mathbf{w})}{\sigma(\mathbf{w})}. \tag{28}$$

From the definition of λ, we also have $\mathrm{SR}(\mathbf{w}) = \lambda \sigma(\mathbf{w})$, so substituting into (28):

$$\lambda \sigma(\mathbf{w}) = \frac{\gamma - r + \lambda \sigma^2(\mathbf{w})}{\sigma(\mathbf{w})} \quad \Rightarrow \quad \lambda \sigma^2(\mathbf{w}) = \gamma - r + \lambda \sigma^2(\mathbf{w}) \quad \Rightarrow \quad \boxed{\gamma = r.} \tag{29}$$

Step 3 — Implied returns

Substituting $\gamma = r$ back into (26):

$$\boldsymbol{\mu} - r\mathbf{1} = \lambda \boldsymbol{\Sigma}\mathbf{w}, \qquad \lambda = \frac{\mathrm{SR}(\mathbf{w})}{\sigma(\mathbf{w})}. \tag{30}$$

This is the *reverse-optimisation* formula: given any portfolio $\mathbf{w}$, we **define** its implied returns as the unique $\tilde{\boldsymbol{\mu}}(\mathbf{w})$ satisfying this FOC:

$$\tilde{\boldsymbol{\mu}}(\mathbf{w}) = r\mathbf{1} + \frac{\mathrm{SR}(\mathbf{w})}{\sigma(\mathbf{w})} \boldsymbol{\Sigma}\mathbf{w}. \tag{31}$$

Step 4 — Implied bet

Applying (31) to both $\mathbf{w}^\star(t)$ and the benchmark $\mathbf{b}$:

$$\tilde{\mu}_i(\mathbf{w}^\star(t)) = r + \frac{\mathrm{SR}}{\sigma(\mathbf{w}^\star(t))} [\boldsymbol{\Sigma}\mathbf{w}^\star(t)]_i, \tag{32}$$

$$\tilde{\mu}_i(\mathbf{b}) = r + \frac{\mathrm{SR}}{\sigma(\mathbf{b})} [\boldsymbol{\Sigma}\mathbf{b}]_i. \tag{33}$$

The implied bet isolates the deviation in sector i's priced return due solely to the ESG constraint:

$$\Delta\tilde{\mu}_i(\mathbf{w}^\star(t)) = \tilde{\mu}_i(\mathbf{w}^\star(t)) - \tilde{\mu}_i(\mathbf{b}) = \frac{\mathrm{SR}}{\sigma(\mathbf{w}^\star(t))} [\boldsymbol{\Sigma}\mathbf{w}^\star(t)]_i - \frac{\mathrm{SR}}{\sigma(\mathbf{b})} [\boldsymbol{\Sigma}\mathbf{b}]_i. \quad \square \tag{34}$$